PEER REVIEW OF THE FAMILY CLASS OVERSEAS SPOUSE OR PARTNERS APPLICATIONS ELIGIBILITY MODELS

EXECUTIVE SUMMARY

Immigration, Refugees and Citizenship Canada
Immigration, Réfugiés et Citoyenneté Canada
Canada

Background

Automated tools are part of Immigration, Refugees and Citizenship Canada (IRCC)'s commitment to using technology responsibly to build a stronger immigration system for all of our clients. Recently, IRCC developed technical systems that streamline eligibility assessments for family class overseas spousal and partner applications (herein referred to as the "FC OVS models"). As required by the Treasury Board Secretariat's (TBS) *Directive on Automated Decision-Making*, a peer review of the models was conducted. For this initiative, the peer review was completed by the Data Science Team at Shared Services Canada (SSC).

The goal of the peer review was to provide an informed assessment of whether the technology is being used responsibly by validating IRCC's *Algorithmic Impact Assessment*, the quality of the system, the appropriateness of the quality assurance and risk mitigation measures, as well as to identify the residual risk of operating the system.

The Family Class Overseas Spouse and Partners Models

One of several immigration objectives laid out in Canada's *Immigration and Refugee Protection Act* is "to see that families are reunited in Canada". In support of this objective, IRCC is expanding its use of advanced analytics to process applications more efficiently and to help applicants living overseas reunite with their spouse or partner in Canada sooner.

Two advanced analytics (AA) models for automating the sponsorship approvals and principal applicant (PA) eligibility approvals of routine applications under the Spouse and Partner categories of the Family Class Overseas were developed. The models are expected to support processing efficiencies, decrease processing times for active eligibility cases, improve workflow management, and enhance strategic resource allocation.

Both models triage applications into two tiers. Tier 1 cases are considered routine and eligibility approvals of Sponsors (SPR) or Principal Applicants (PA) can be automated using the Sponsor model or the Principal Applicant model. Applications that do not have principal applicant eligibility or sponsorship approved by the models (Tier 2) are triaged to officers for regular manual processing as per standard procedures. The tools do not refuse or recommend refusing applications, and the final decision on all applications is always made by an IRCC officer.

Peer Review by SSC

SSC conducted a peer review to assess IRCC advanced analytics solution for spouse and partners under the Family Reunification Program. This review is mandated by the TBS to ensure compliance with its policy direction on artificial intelligence applications. The peer review assesses IRCC's machine learning models that sort and help process spouse and partners applications in the overseas family class. The peer review evaluates the compliance, impact, transparency, accountability, ethical and responsible use, training, and improvement areas of the IRCC's solution, based on the TBS Directive on Automated Decision-Making. A report is produced at the end of the peer review to describe the process, the findings, and the recommendations of the peer reviewer.

During the peer review process, the Shared Services Canada (SSC) team, comprised of data scientists and business analysts, worked in tandem with IRCC's essential stakeholders and domain specialists to gain a deeper comprehension of the intricate details of the advanced analytics solution, including its supporting infrastructure and processes. To guide the assessment process and ensure SSC was following available best practices, the SSC team referred to the Bronson/Millar "Peer Review for Automated Decision-Making Tools Under Canada's Directive on Automated Decision-Making – A Model Peer Review Process and Recommendations for Best Practices".

The report concludes that, based on the provided artefacts and interviews, the IRCC's solution is robust, and necessary controls and safeguards have been implemented to ensure client privacy is safeguarded throughout its lifecycle. The IRCC team has sufficiently met the requirements outlined in the TBS Algorithmic Impact Assessment and in accordance with the TBS's Directive on Automated Decision-Making.

SSC's Recommendations and IRCC's Responses

Recommendation: Training is provided for officers; however, the adequacy of the training may need to be re-evaluated and updated once the algorithms have been used in a production setting for a period of time. Periodic re-evaluation of employee training to ensure the training material remains relevant and up to date.

IRCC Response: Officers are provided with training on IRCC's automated tools to ensure they understand that a lack of an automated approval does not constitute a recommendation to refuse an application. IRCC also conducts regular quality assurance exercise to ensure that officer decisions are not impacted by automated approvals.

An ongoing feedback mechanism was also established so that officers have the opportunity to report issues and suggest improvements.

IRCC will continue to evaluate employee training to ensure the training material remains relevant and up to date.

Recommendation : During the peer review assessment, the mandatory Gender Based Assessment Plus (GBA+) was not available for review. Although the SSC peer reviewers did not evaluate the results and recommendations GBA+ assessment, IRCC has informed the peer reviewers of having completed the GBA+ as per the AIA. Furthermore, IRCC has expressed their intent to proactively implement the necessary controls and safeguards ensuring the advanced analytics solution does not discriminate against marginalized or special groups of individuals.

IRCC Response: In line with the requirements of the TBS Directive on Automated Decision-Making, IRCC completed a GBA+ assessment prior to the launch of the model, and will proactively implement the necessary controls or safeguards.

The models will be monitored and undergo regular quality assurance exercises to test their accuracy and ensure they continue to function as intended. Following the implementation of the models, periodic GBA Plus assessments will also be conducted once a sufficient number of applications have been processed with the models. This will allow an assessment of potential differential impacts or unintended outcomes over time.